Instance Segmentation

Note: If using the docker container from the factory image, you don't need to rebuild the environment. The environment is already set up, you just need to enter the docker according to the previous tutorial and run the corresponding function commands to use it.

1. Instance Segmentation: Image

Use yolo11n-seg.pt to predict images that come with the ultralytics project.

Enter the code folder:

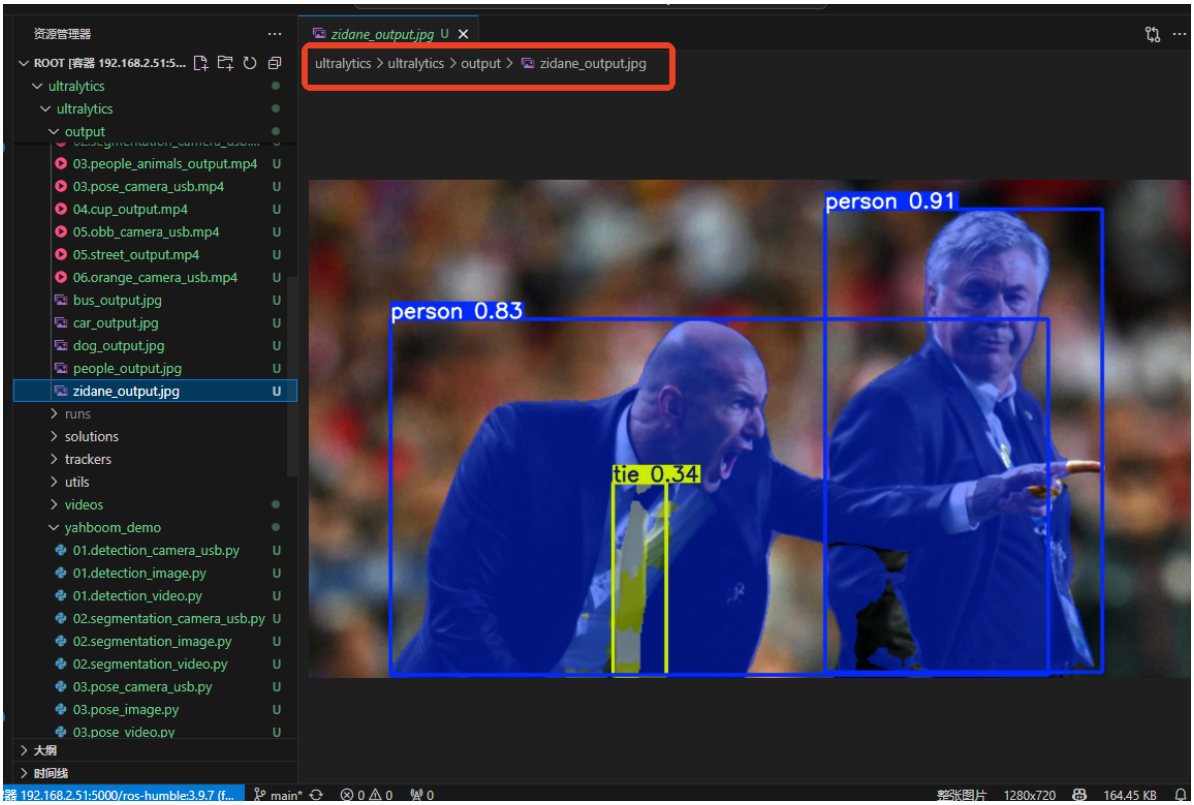
```
cd ~/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 02.segmentation_image.py
```

Effect Preview

yolo recognition output image location: /root/ultralytics/ultralytics/output/



Example code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg.pt")  # pretrained
yolo11n-seg model
# model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg.onnx")
# model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg_ncnn_model")
```

```
# Run batched inference on a list of images
results = model("/root/ultralytics/ultralytics/assets/zidane.jpg")  # return a
list of Results objects

# Process results list
for result in results:
    # boxes = result.boxes  # Boxes object for bounding box outputs
    masks = result.masks  # Masks object for segmentation masks outputs
    # keypoints = result.keypoints  # Keypoints object for pose outputs
    # probs = result.probs  # Probs object for classification outputs
    # obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen

result.save(filename="/root/ultralytics/ultralytics/output/zidane_output.jpg")  #
save to disk
```

2. Instance Segmentation: Video

Use yolo11n-seg.pt to predict videos in the ultralytics project (not videos that come with ultralytics).

Enter the code folder:

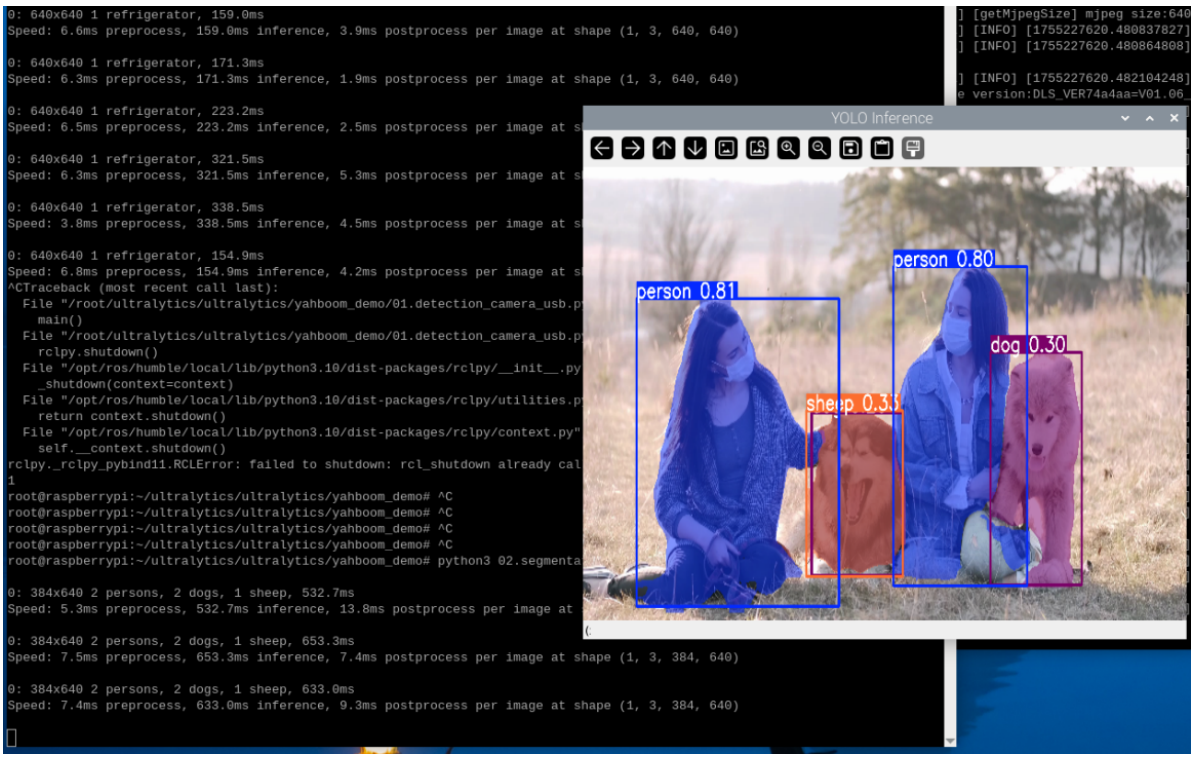
```
cd /root/ultralytics/ultralytics/yahboom_demo
```

Run the code:

```
python3 02.segmentation_video.py
```

Effect Preview

yolo recognition output video location: /root/ultralytics/ultralytics/output/



Example code:

```
import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg.pt")
# model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg.onnx")
# model = YOLO("/root/ultralytics/ultralytics/yolo11n-seg_ncnn_model")

# Open the video file
video_path = "/root/ultralytics/ultralytics/videos/people_animals.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a VideoWriter object to output the processed video
output_path =
"/root/ultralytics/ultralytics/output/02.people_animals_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v')  # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break
# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()
```

3. Instance Segmentation: Real-time Detection

3.1 Start Camera

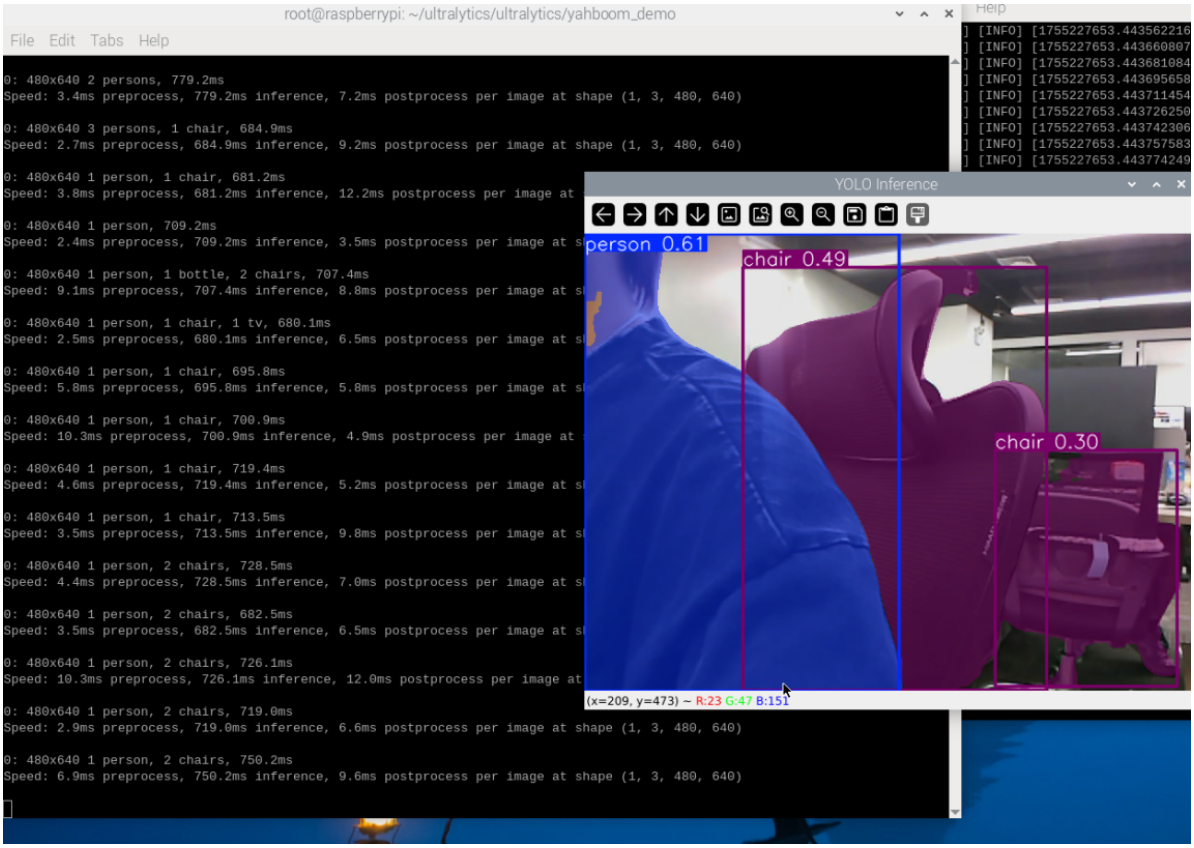
Start the following program according to your camera model, enter in terminal:

```
ros2 launch dofbot_info camera_arm_kin.launch.py
```

Open another terminal and enter the code folder:

```
cd /root/ultralytics/ultralytics/yahboom_demo
```

Run the code: Click on the preview screen and press q key to terminate the program!

```
python3 02.segmentation_camera_usb.py
```

Effect Preview